

Requirement Specification Document

Smart Renewable Energy Management System

Week 1 – Requirement Analysis and Customer Database Module

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Week	Week 1

1.Introduction

The **Smart Renewable Energy Management System** is designed to monitor, manage, and optimize renewable energy resources such as solar and wind power. The system helps users track energy production and consumption, reduce electricity wastage, and promote sustainable energy usage through a centralized database.

1.1 Purpose

The purpose of this document is to identify and define the business, functional, and database requirements of the Smart Renewable Energy Management System. It provides the foundation for designing the customer database and the overall system.

1.2 Project Scope

The system stores customer information, renewable energy source details, energy consumption records, and energy generation data. It enables users to monitor energy usage, receive usage reports, and helps administrators manage customers and renewable energy devices efficiently.

1.3 Intended Users

- Customers – Monitor their energy usage and renewable energy production.
- Administrators – Manage customer accounts, energy devices, and reports.
- Energy Providers – Monitor renewable energy generation and system performance.

2. Business Requirements

The system should:

- Monitor renewable energy generation in real time.
- Record and analyze electricity consumption.
- Reduce unnecessary energy wastage.
- Generate daily, weekly, and monthly energy reports.
- Notify users about high energy consumption.
- Support sustainable and cost-effective energy management.

3. Functional Requirements

The system must:

- Maintain customer information securely.
- Record renewable energy generation and electricity consumption.
- Monitor energy usage in real time.
- Generate daily, weekly, and monthly energy reports.
- Send alerts when energy consumption exceeds predefined limits.
- Support efficient and sustainable energy management.

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow customers to register and log in securely.
FR-2	The system shall store customer profile information.
FR-3	The system shall record renewable energy generation data.
FR-4	The system shall monitor electricity consumption.
FR-5	The system shall generate energy usage reports.
FR-6	The system shall send notifications for high energy consumption.
FR-7	The system shall allow administrators to manage customer and device information.
FR-8	The system shall display dashboards showing energy production and consumption.

4. Non-Functional Requirements

- **Performance:** The system should retrieve energy data within 2 seconds.
- **Security:** Customer information and energy records must be protected.
- **Reliability:** The system should provide accurate and consistent energy data.
- **Usability:** The interface should be easy for both customers and administrators.
- **Scalability:** The database should support an increasing number of customers and energy devices.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
customer_id	INT (Primary Key)	Unique customer identifier
full_name	VARCHAR(100)	Customer full name
email	VARCHAR(100)	Customer email address
phone_number	VARCHAR(15)	Customer contact number
address	VARCHAR(255)	Customer address
energy_source	VARCHAR(50)	Solar, Wind, or Hybrid

Attribute	Data Type	Description
installation_date	DATE	Renewable energy installation date
registration_date	DATE	Customer registration date

6. Assumptions and Constraints

- The system uses a relational database such as MySQL or PostgreSQL.
- Only registered customers can access the system.
- Internet connectivity is required for real-time monitoring.
- Renewable energy devices (such as solar panels or wind turbines) are installed and connected.
- This document covers **Week 1 – Requirement Analysis and Customer Database Module** only.

7. Conclusion

This Requirement Specification Document defines the initial business, functional, and database requirements for the **Smart Renewable Energy Management System**. The Customer Database Module provides the foundation for designing the database schema, ER diagram, and future system implementation.